ASC Database User's Manual

Brent Young

March 24, 2026

Contents

Contents i

List of Figures iii

Notice v

Developer Contact vi

1 About Ubuntu 1

2 About MySQL 1

3 About Python 3

4 About LaTeX 4

5 Intro to the ASC–DB 7

6 Reports 9
- 6.1 Date Range Report . . . 9
- 6.2 Specific WINs Report . . . 11
- 6.3 View by WIN . . . 14
- 6.4 DB Statistics . . . 15

7 DB Management 17
- 7.1 Add Single Entry . . . 17
- 7.2 Modify Existing Entry . . . 19
- 7.3 Add Entries from File . . . 21
- 7.4 Delete Entries by WIN . . . 27

7.5 Delete Entries by Date 28

8 Backup and Restore **30**

8.1 Backup Current DB 30

8.2 Restore DB from File 31

List of Figures

1.0.1 Software Updater Icon 2

2.0.1 Description of `Appeal_Records` 1
2.0.2 Terminal Icon 2

3.0.1 ASC Database Interact Icon 3

4.0.1 Sample LaTeX Markup Code vs. Rendered Document 4
4.0.2 TeXStudio Icon 5

5.0.1 ASC Database Interact Icon 7
5.0.2 ASC Database Log In Prompt 7
5.0.3 ASC Database Main Control 8

6.1.1 Date Range Prompt 9
6.1.2 Date Range File Prompt 10
6.1.3 Date Range Directory Prompt 10
6.1.4 Date Range Success Notification 11
6.2.1 WIN Entry Dialogue 12
6.2.2 Specific WINS Report File Prompt 12
6.2.3 Specific WINS Report Directory Prompt 13
6.2.4 Specific WINS Report Success Notification 13
6.3.1 View by WIN Prompt 14
6.3.2 View by WIN Tabbed Window 15
6.4.1 DB Statistics Pop Up Window 16

7.1.1 New Entry Window 17
7.1.2 New Entry Verification Window 18
7.1.3 Add Entry Success Notification 19

7.2.1 Modify Existing Entry Prompt 19
7.2.2 Entry Selection Window 20
7.2.3 Modify Existing Entry Window 20
7.2.4 Modify Entry Success Notification 21
7.3.1 `LibreOffice Calc` Icon 21
7.3.2 Sample Spreadsheet with Errors 22
7.3.3 Spreadsheet `Save As` Settings 23
7.3.4 CSV Options 24
7.3.5 CSV File Formatting Reminder 25
7.3.6 File Prompt 25
7.3.7 Add Entries from File Result Notification 26
7.3.8 Add Entries from File Error Details 26
7.4.1 WIN Entry Dialogue 27
7.4.2 Delete by WIN Warning 28
7.4.3 Delete by WIN Final Notification 28
7.5.1 Delete by Date Prompt 29
7.5.2 Delete by Date Warning 29
7.5.3 Delete by Date Final Notification 29

8.1.1 Back Up Directory Prompt 31
8.2.1 Restore DB Directory Prompt 31
8.2.2 Restore DB Warning 32

Notice

The information contained in the Academic Standards Committee Database of Appeal Records (ASC–DB) is highly confidential. The information contained in the database should only be used by duly designated members of the Wilkes University Academic Standards Committee and only for purposes officially sanctioned by the committee.

Developer Contact

Brent Young
Associate Professor of Mathematics
Wilkes University
Office: SLC 406
Email: brent.young@wilkes.edu
Ext: 3824

Chapter 1

About Ubuntu

The machine currently housing the Academic Standards Committee Database of Appeal Records (ASC–DB) uses the operating system Ubuntu which is an open–source OS based on Linux. As of this writing, the version of Ubuntu in use is recorded below.

- Version: Ubuntu 20.04.4 LTS
- Codename: Focal Fossa

Support for this version of Ubuntu is guaranteed through April 2025 (and can most likely be used well past that date). Procedures to upgrade Ubuntu to the latest version can be found on the internet. The most reliable source would be Ubuntu's own reference website:
https://ubuntu.com/tutorials/upgrading-ubuntu-desktop
Be aware that upgrades can take a considerable length of time! Feel free to contact the software developer (Brent Young: brent.young@wilkes.edu) if you think that an upgrade to the latest version of Ubuntu is necessary.

Windows and Mac users should have few problems getting used to Ubuntu's window manager. This should especially be true since the computer is likely only to be used to run the ASC–DB. Most of the applications you are likely to need have been added to the Favorites Bar on the left side of the desktop. Other applications can be accessed by clicking on the grid of 9 squares located at the bottom of the Favorites Bar. It should be noted that the standard productivity software loaded with Ubuntu is the LibreOffice suite of applications. These function similarly to Microsoft Office products, but there are occasional formatting differences when opening documents prepared by one

family of programs with the other. This should not be an issue with PDF documents.

It is recommended that you run the Software Updater application periodically. This will keep all applications up to date as well as provide security updates to the operating system. To run an update, simply click the following icon located on the Favorites Bar. If it has been awhile since the last update, this may require a good amount of time!

Figure 1.0.1: Software Updater Icon

Chapter 2

About MySQL

The Academic Standards Committee Database of Appeal Records (ASC–DB) is implemented using MySQL, an open–source relational database management system first released in 2000. The database is named `Appeals_DB` and contains precisely one table: `Appeal_Records`. Each entry in `Appeal_Records` consists of 11 fields.

FIELD	TYPE	Required?	Description
`WIN`	Unsigned Integer	Yes	Wilkes Identification Number
`First_Name`	String	Yes	Student first name
`Last_Name`	String	Yes	Student last name
`Date`	Date String	Yes	Appeal Date: YYYY-MM-DD
`Time`	String	No	Appeal Time: 24 Hour Format
`Committee_Members`	Mediumtext	No	List of Committee Members
`Note_Taker`	String	No	Initials of Note Taker
`Minutes`	Longtext	Yes	Minutes of appeal
`Motion`	Mediumtext	No	Motion on appeal
`Decision`	Mediumtext	No	Committee decision on appeal
`Readmit`	Character	Yes	Single Character: "Y" or "N"

Figure 2.0.1: Description of `Appeal_Records`

The fields `WIN` and `Date` are combined to form the primary key for each entry. This means that MySQL distinguishes between different entries by requiring entries to have a *unique* combination of WIN and appeal date. As

a consequence, there can only be one entry for any single WIN on any given appeal day. If for some unforeseen reason a student appears more than once on a given appeal day, the minutes for that student on that particular day should be combined into a single entry. Otherwise, MySQL will give an error and the data will likely be lost!

Users can interact directly with the ASC–DB through MySQL itself. As currently implemented, however, this requires using the command line interface. To begin, click on the terminal icon in the Favorites Dock in the upper left–hand–side of the desktop. After the terminal opens, access MySQL by

Figure 2.0.2: Terminal Icon

typing `mysql -u root -p` on the command line. After successfully entering the password, typing `USE Appeals_DB;` on the command line will connect you to the ASC–DB.

Unfortunately, this manual is not the place to dive into the details of using Structured Query Language (SQL). For reference, see *Murach's MySQL (2e)* by Joel Murach (ISBN: 978–1–890774–82–0). Hopefully, the ASC Database Interact will be sufficient for most uses of the ASC–DB!

If for some reason the entire Appeals_DB database is dropped, then the ASC Database Interact may not be able to restore the program properly. In such case, contact the developer (Brent Young: brent.young@wilkes.edu) who should be able to help set things back up correctly.

Chapter 3

About Python

The ASC Database Interact GUI (graphical user interface) is written in the programming language Python. As of this writing, the version of Python in use was 3.8.10. The GUI has been set up so that interacting with Python itself should not be necessary. Simply clicking on the ASC Database icon should start up the graphical user interface without having to run the program through the Python interpreter.

Figure 3.0.1: ASC Database Interact Icon

If the machine housing the ASC Database is reformatted or the database is moved to a different machine, there are several packages that the GUI relies on to function properly. A list of all required packages can be found at the beginning of the file `ASC_DB_Interact.py`. The two packages which are mostly likely to need separate installation from Python itself are the following.

- `tkinter` (TK Interface which manages all pop-up windows for the GUI)
- `mysql.connector` (which handles communications between Python and MySQL)

Chapter 4

About LaTeX

```
By ``quantizing" the classical
angular momentum (i.e. setting
$\mathbf{p}=-i\hbar\nabla)$,
we get the Orbital Angular
Momentum Operator
$$ \mathbf{L} = \mathbf{r}
\times \frac{\hbar}{i}
\mathbf{\nabla} $$
whose three components
\begin{align*}
L_x &= \frac{\hbar}{i}
(y\partial_z-z\partial_y)\\
L_y &= \frac{\hbar}{i}
(z\partial_x-x\partial_z)\\
L_z &= \frac{\hbar}{i}
(x\partial_y-y\partial_x)
\end{align*}
are well--defined self--adjoint
operators.
```

By "quantizing" the classical angular momentum (i.e. setting $\mathbf{p}=-i\hbar\nabla$), we get the Orbital Angular Momentum Operator

$$\mathbf{L} = \mathbf{r} \times \frac{\hbar}{i}\nabla$$

whose three components

$$\begin{aligned}
L_x &= \frac{\hbar}{i}(y\partial_z-z\partial_y)\\
L_y &= \frac{\hbar}{i}(z\partial_x-x\partial_z)\\
L_z &= \frac{\hbar}{i}(x\partial_y-y\partial_x)
\end{aligned}$$

are well–defined self–adjoint operators.

Figure 4.0.1: Sample LaTeX Markup Code vs. Rendered Document

LaTeX(pronounced "LAY–tekh" or "LAH–tekh") is a markup language in

which documents are prepared by typing content to be displayed along with various formatting commands.

Preparing documents with LaTeX [1] takes considerably more effort than the common what–you–see–is–what–you–get word processors like Microsoft Word and Apple Pages. Fortunately, the ASC–DB Interact handles most of the work for the user.

The exact details of how to produce LaTeX files are mostly unimportant here. All that matters is that some version of LaTeX be installed on the laptop housing the ASC–DB. Since the laptop uses Ubuntu for its operating system, LaTeX was installed through TeX Live (TeX version 3.14159265 as of June 2019). The status of the LaTeX installation can be checked by entering the command `tex --version` on the command line.

In the off chance that you need to edit a LaTeX file manually, the TeX editor TeXStudio has been installed on the laptop housing the ASC–DB. Even though it is perfectly possible to write a `tex` file in most any text editor,

Figure 4.0.2: TeXStudio Icon

TeXStudio has a lot of features that make typing LaTeX much more efficient – including delimiter matching and auto–completion of LaTeX commands.

Anytime that a report is generated by the ASC Database Interact program, a LaTeX file is created and then compiled into a PDF document. You may notice several files pop up in the directory where the report is saved during the process of compilation. Most of these files should be eliminated once the processing of the report is finished. However, the original LaTeX file (having a `.tex` extension) will be left along with the finalized PDF file. The reason for this choice is that on rare occasions some error in the compilation process prevents the creation of the final PDF file. In that event, you can open the LaTeX file in TeXStudio, press the double green arrow icon near the top of the editor, and read the resulting errors in the messages dialog at the bottom of the window. The most likely error is that some unexpected

[1] LaTeX was developed in 1983 by Leslie Lamport as a set of macros inside the TeX typesetting system. TeX was developed in 1978 by Donald Knuth.

unicode character has made its way into the database, and LaTeX is uncertain what to do with it. This particular issue has been much improved in recent versions of TeXLive. If a final PDF document of the report has been created, you can simply discard the unnecessary LaTeX file.

Note: The ASC Database User's Manual itself was written in LaTeX. If you need to edit this document, find the file `ASCDatabase_ReadMe.tex`. It should automatically open in TeXStudio.

Chapter 5

Introduction to the ASC Database Interact

The ASC Database of Appeal Records (ASC–DB) can be started by clicking the icon below from the Favorites Bar to the left of the home screen.

Figure 5.0.1: ASC Database Interact Icon

Clicking this icon should bring up the log in routine.

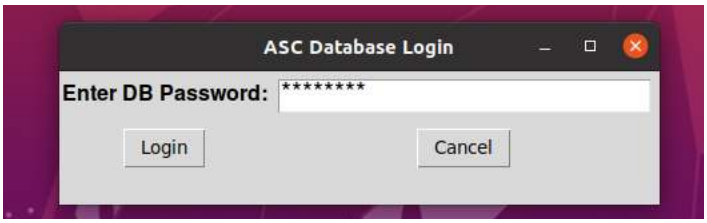


Figure 5.0.2: ASC Database Log In Prompt

Once you have entered the correct password, you will be taken to the main database control window.

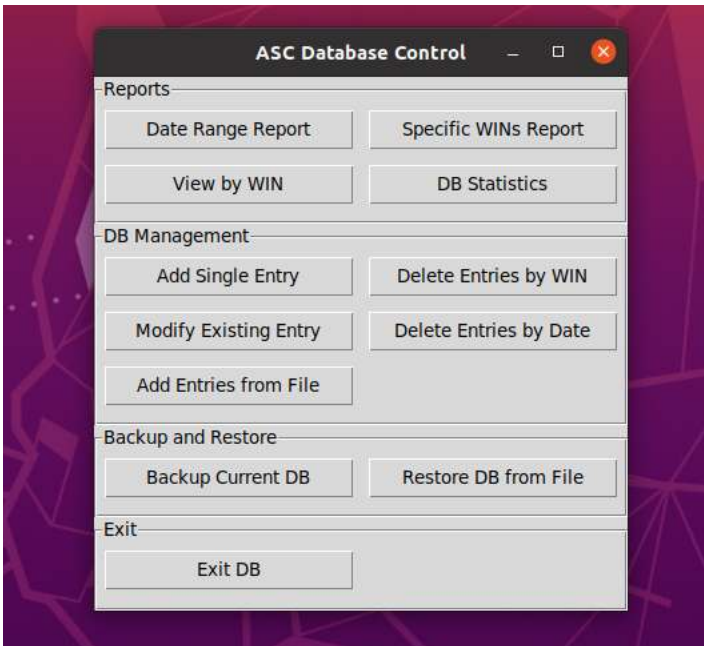


Figure 5.0.3: ASC Database Main Control

The various options in the main control window are arranged into three major groups: Reports, DB Management, and Backup and Restore. Be sure to choose `Exit DB` from the bottom of the window when you are done using the program.

For future reference, the Python code for the database interaction program is stored in the following location.

`/usr/local/ASC_DB/ASC_DB_Interact.py`

The file which sets up the desktop icon for the database interaction program is stored in the following location.

`/usr/share/applications/ASC_DB.desktop`

These files cannot be altered without root privileges.

Chapter 6

Reports

6.1 Date Range Report

As the name suggests, a Date Range Report prints out all entries in the ASC–DB between two given dates. Once selected, the main database control program will bring up a dialogue box asking for the range of dates to search. Notice that dates must be entered in the format `YYYY-MM-DD`.

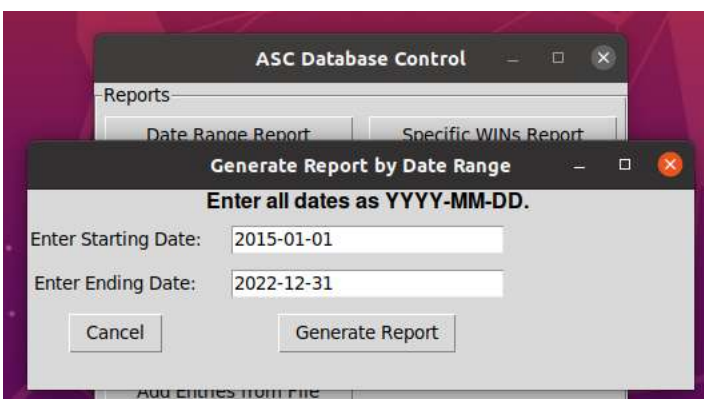


Figure 6.1.1: Date Range Prompt

The program has been designed to catch most common errors (e.g. dates in the wrong order, day out of range of the given month, etc.) but care should be taken.

Once the range of dates has been entered, selecting `Generate Report`

will bring up a file name dialogue. You should enter the file name of the

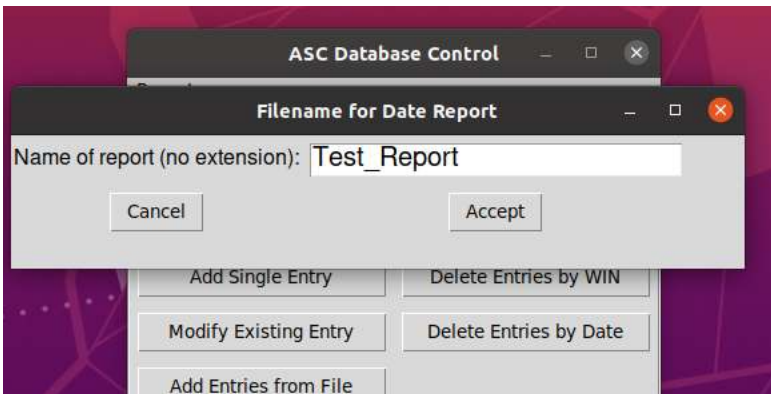


Figure 6.1.2: Date Range File Prompt

report *without giving an extension*. The main database control program will eventually produce a PDF file with the file name you have specified, but there are a number of intermediate files that will be generated first.

After entering the desired file name, a separate dialogue box will open asking you to specify the directory where you would like the final report stored. Once you select `Ok`, a PDF file with the given file name will be

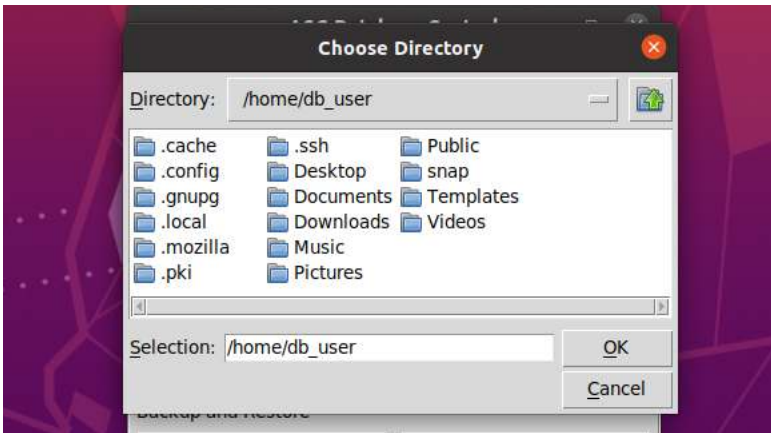


Figure 6.1.3: Date Range Directory Prompt

created in the directory you have specified. An auxiliary LaTeX file will also

be created (which can be discarded assuming the PDF report was generated successfully.) Once the program has finished producing the report, a prompt will appear reminding you of the location of the report.

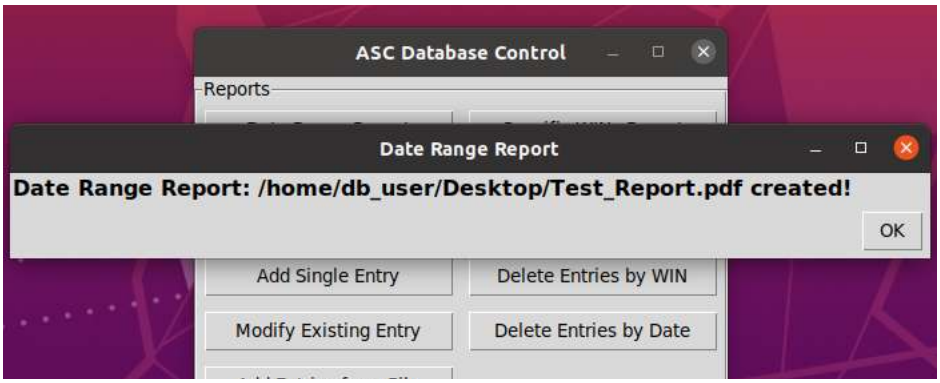


Figure 6.1.4: Date Range Success Notification

6.2 Specific WINs Report

The Specific WINs Report complies a PDF document consisting of all database entries associated to the Wilkes Identification Numbers (WINs) provided by the user. The database entries are arranged in the order that the WINs were supplied by the user, and a cover page is generated giving student name, WIN, and number of entries for each student in the report. If a WIN supplied by the user has no associated entries, it is skipped and no record for it is noted in the final report.

Selecting `Specific WINs Report` from the main control window will bring up the WIN Entry dialogue box. Each WIN should be entered *without dashes*, and a single comma should separate each WIN (no comma is necessary after the last WIN). *DO NOT PRESS ENTER AFTER ANY WIN.* The window has been sized to make each WIN appear on its own line for ease of reading. After all WINs have been entered, clicking `Generate Report` will begin the process of compiling the PDF document.

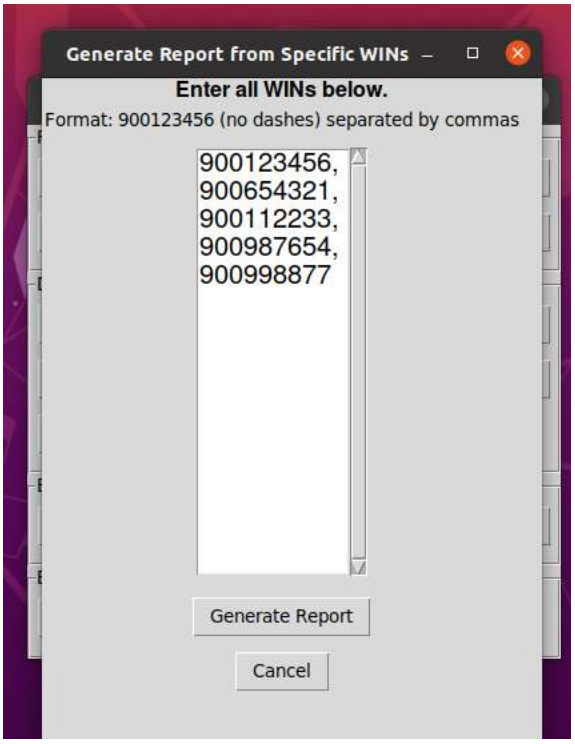


Figure 6.2.1: WIN Entry Dialogue

First, a prompt will appear asking for the file name of the report. You

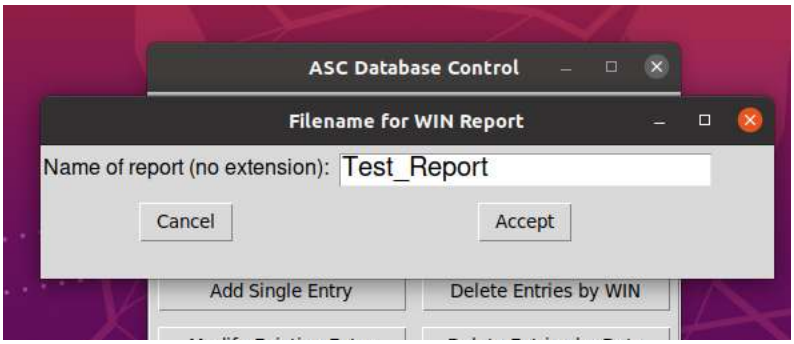


Figure 6.2.2: Specific WINS Report File Prompt

should enter the file name of the report *without giving an extension*. The

main database control program will eventually produce a PDF file with the file name you have specified, but there are a number of intermediate files that will be generated first.

After entering the desired file name, a separate dialogue box will open asking you to specify the directory where you would like the final report stored. Once you select `Ok`, a PDF file with the given file name will be

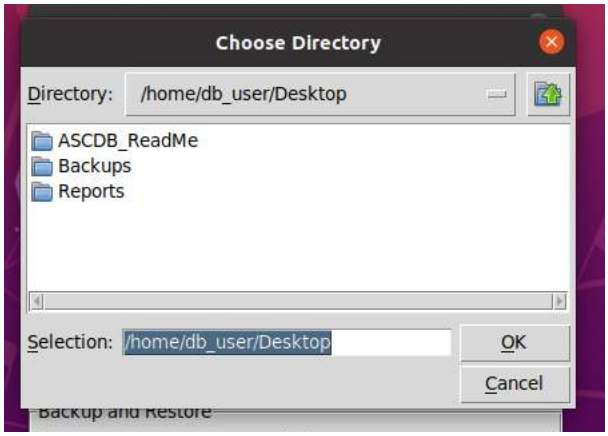


Figure 6.2.3: Specific WINS Report Directory Prompt

created in the directory you have specified. An auxiliary LaTeX file will also be created (which can be discarded assuming the PDF report was generated successfully.) Once the program has finished producing the report, a prompt will appear reminding you of the location of the report.

Figure 6.2.4: Specific WINS Report Success Notification

6.3 View by WIN

Frequently, a student will request an appeals hearing at the last minute. The View by WIN report option allows the user to view all entries associated to a student's WIN without having to generate an actual report file. This can be very useful in case the Specific WINs Report was generated before the student requested a hearing.

Selecting the `View by WIN` option from the main control window will bring up a dialogue asking for a single WIN. The WIN for the student in question should be typed *without dashes*. Select `View` once the desired WIN

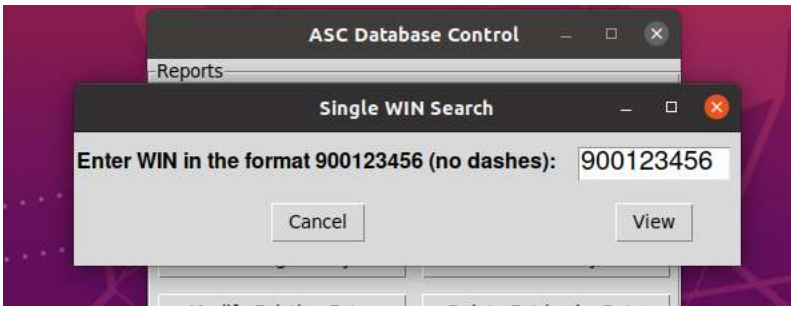


Figure 6.3.1: View by WIN Prompt

has been entered. If there are no entries associated to the given WIN, a dialogue box will be shown to announce that fact. Otherwise, a tabbed window will appear showing all entries matching the given WIN. Each tab is headed by the date and time of the appeal hearing shown. Selecting `Close` at the bottom of the window will return you to the main database control window.

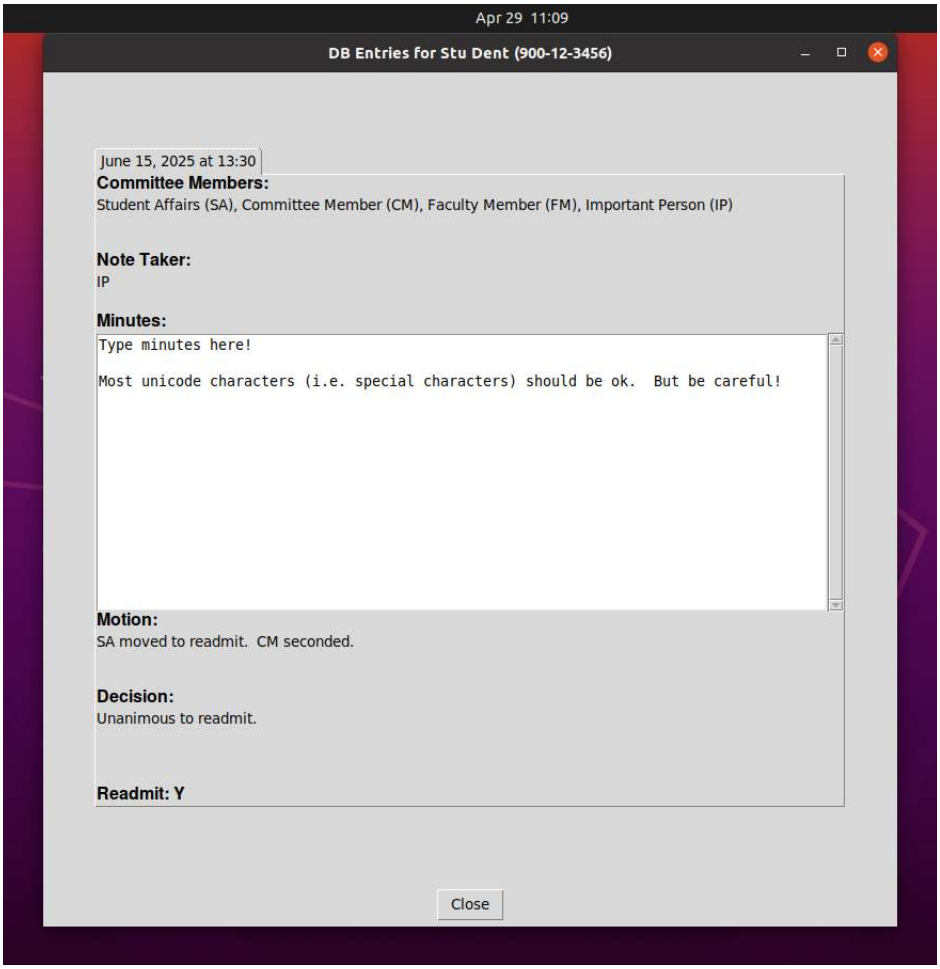


Figure 6.3.2: View by WIN Tabbed Window

6.4 DB Statistics

Clicking `DB Statistics` from the main control window will bring up a small pop–up window with the following information about the current database.

- Total number of entries in the database

- Earliest appeal date in the database
- Latest appeal date in the database
- Number of distinct WINs in the database
- Number of successful appeals
- Number of denied appeals

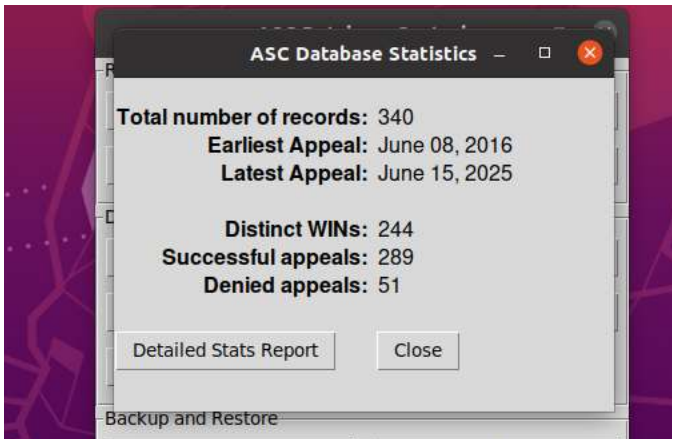


Figure 6.4.1: DB Statistics Pop Up Window

Selecting `Close` will return you to the main database control window.

Selecting `Detailed Stats Report` will print out a PDF file with more details about the contents of the ASC Database. In addition to the basic information given in the pop up window, the report will break down appeals by number of first-time, second-time, third-time, and fourth-time and higher appeals. It will also give the relative percentages of successful appeals in each of those categories.

Before the detailed report is created, the user will be prompted to specify the directory where the report will be saved. Once selected, a PDF report will be generated. Unlike other reports, the auxiliary LaTeX file will be discarded automatically. Finally, a dialogue box giving the name and location of the detailed report will pop up. The file name for the report will be of the form `AppealsDB_Stats_date.pdf` where the `date` field will be replaced by the date the report was generated.

Chapter 7

DB Management

7.1 Add Single Entry

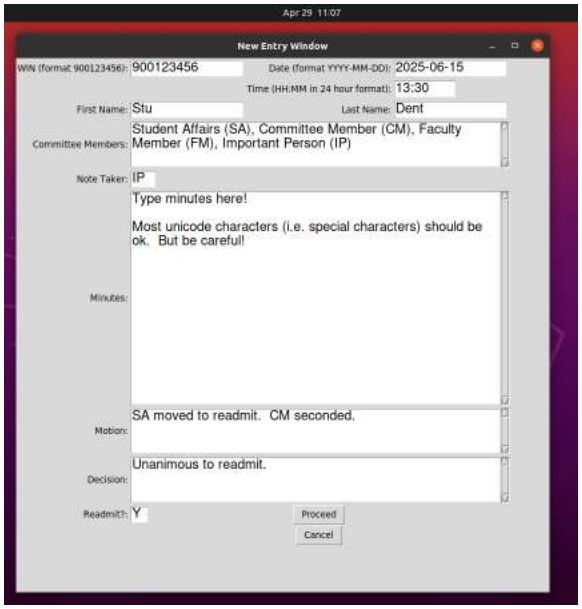


Figure 7.1.1: New Entry Window

Adding entries to the database is one of the most important routine tasks

carried out by the administrator of the ASC Database. There are two main ways to add entries to the database, and the first is `Add Single Entry`. This option is useful if a student appeal is considered after the regular appeals day(s) and the bulk of appeals have already been entered. When entering a large number of appeals at the same time, the `Add Entries from File` option will likely be more efficient.

Selecting `Add Single Entry` from the main database control window will bring up a new entry window. Remember that `WIN, Date, First_Name, Last_Name, Minutes,` and the `Readmit` fields are required. Clicking on `Proceed` will bring up a verification window. Any of the required fields

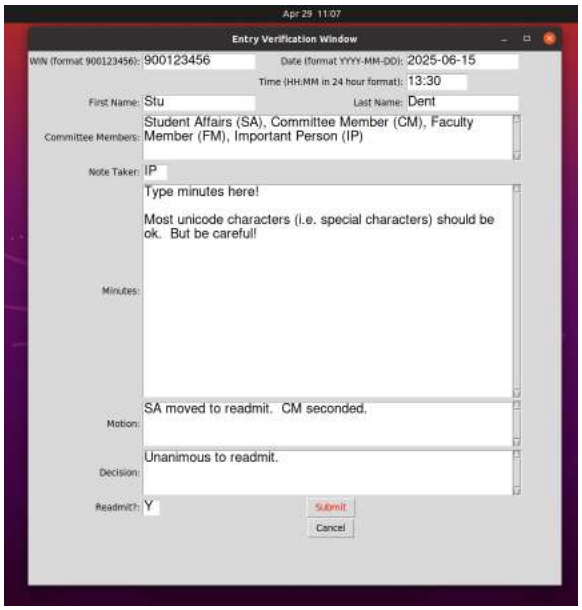


Figure 7.1.2: New Entry Verification Window

which were left blank or improperly formatted will be highlighted in red on the verification screen. *The program will not allow a user to submit an entry if any required field does not meet its specification.* Once the entry has been verified, clicking the `Submit` button will add the entry to the database, and a notification dialogue will appear confirming that fact.

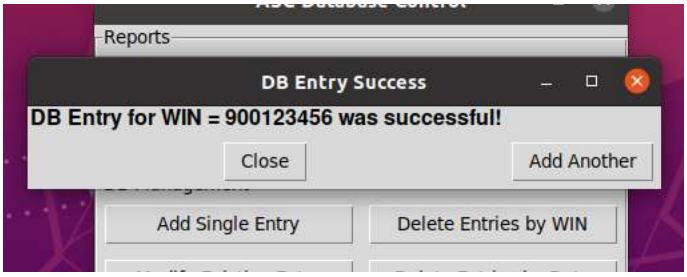


Figure 7.1.3: Add Entry Success Notification

7.2 Modify Existing Entry

If an existing entry needs to be corrected or amended, select the `Modify Existing Entry` option from the main control window. The user will then be asked to supply the WIN for the entry that needs updating. If there is

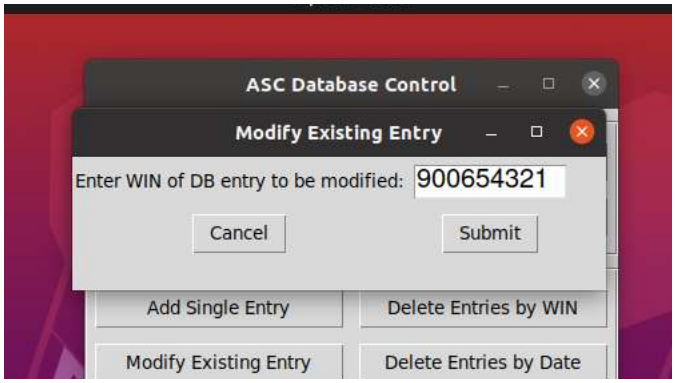


Figure 7.2.1: Modify Existing Entry Prompt

only one entry matching the given WIN, the user will directly be taken to a screen where the existing entry can modified. Otherwise, a dialogue box listing all dates associated to the given WIN will appear. Once the user selects the appropriate entry, a window will appear where the current data for the entry can be modified. *Be aware that the program will not allow a user to submit an entry if any required field does is not properly formatted.* Once the modifications are complete, selecting `Submit` will cause the existing entry to be replaced with the modified version. This process is irreversible

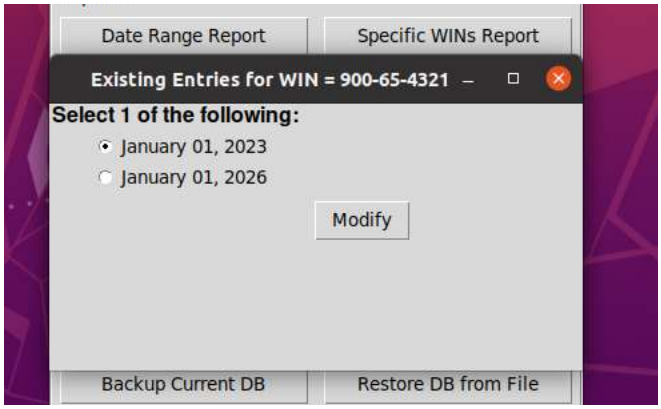


Figure 7.2.2: Entry Selection Window

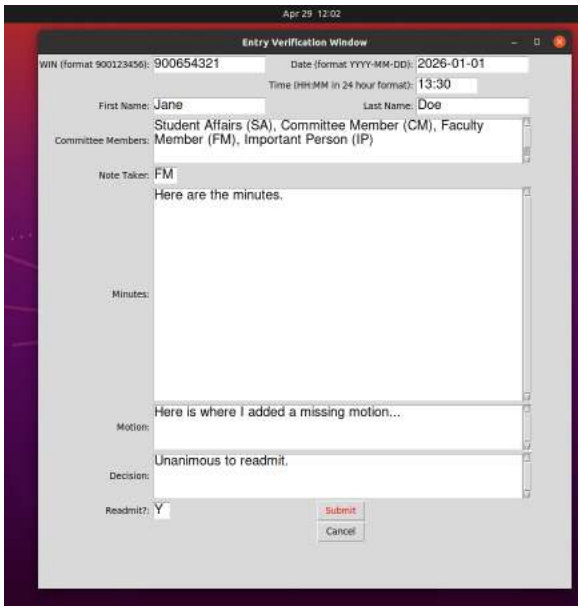


Figure 7.2.3: Modify Existing Entry Window

unless a backup file with the original entry is available. Once complete, the user will be notified that the modification was successful.

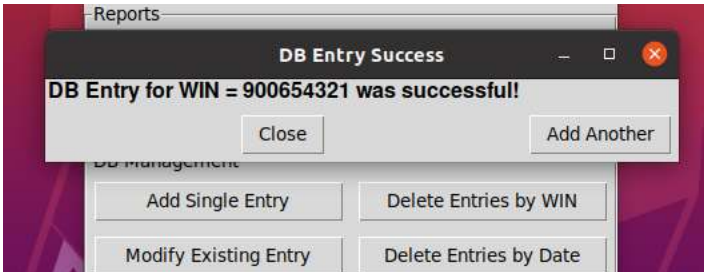


Figure 7.2.4: Modify Entry Success Notification

7.3 Add Entries from File

If you have a relatively large number of appeals to enter at one time, it may be quicker to first prepare a spreadsheet with each entry as a single row. This is especially true if you have a large number of appeals from the same date (as the date and list of committee members will be the same for all entries). It is probably a good idea to note the total number of entries in the database (using the `DB Statistics` option) before adding entries en masse. Running the same check at the end of the process is a simple way to ensure that all entries have been successfully added.

The laptop housing the ASC–DB comes equipped with an open source spreadsheet program called `LibreOffice Calc`. This spreadsheet program should be easy to use if you are familiar with other spreadsheet software (like `Microsoft Excel`).

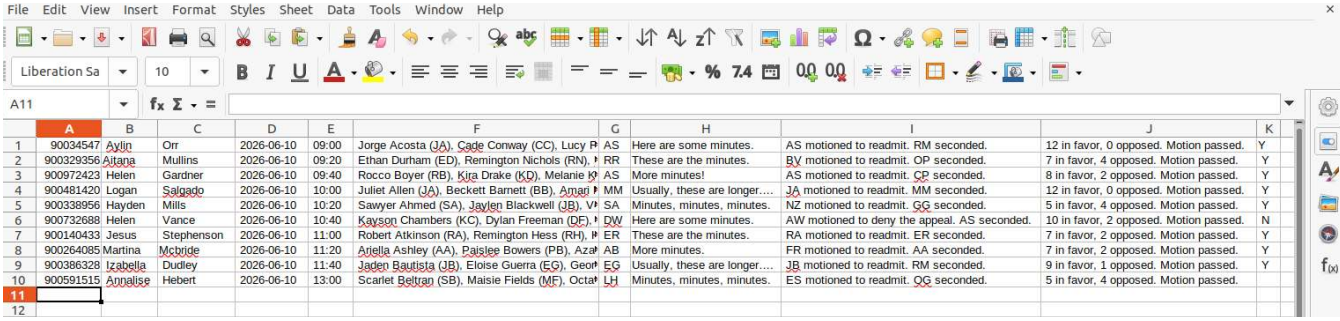
Figure 7.3.1: `LibreOffice Calc` Icon

In order for the entries to be read into the ASC–DB Interact correctly,

you must get the file in the appropriate format. First, each row of the spreadsheet should represent a single appeal record. The fields in each row must be entered in the following order:

1. `WIN`: 9 digit integer (no dashes)
2. `First_Name`
3. `Last_Name`
4. `Date`: in the format YYYY-MM-DD
5. `Time`: HH:MM in 24 hour format
6. `Committee_Members`
7. `Note_Taker`
8. `Minutes`
9. `Motion`
10. `Decision`
11. `Readmit`: single character "Y" or "N"

A sample spreadsheet is shown below. Notice that the first entry has an improper WIN while the last entry has no Readmit entry. This was done intentionally to demonstrate the ability of the program to catch basic errors.



	A	B	C	D	E	F	G	H	I	J	K
1	90034547	Aylin	Orr	2026-06-10	09:00	Jorge Acosta (JA), Cade Conway (CC), Lucy P	AS	Here are some minutes.	AS motioned to readmit. RM seconded.	12 in favor, 0 opposed. Motion passed.	Y
2	900329356	Aitana	Mullins	2026-06-10	09:20	Ethan Durham (ED), Remington Nichols (RN),	RR	These are the minutes.	BV motioned to readmit. OP seconded.	7 in favor, 4 opposed. Motion passed.	Y
3	900972423	Helen	Gardner	2026-06-10	09:40	Rocco Boyer (RB), Kira Drake (KD), Melanie K	AS	More minutes!	AS motioned to readmit. CP seconded.	8 in favor, 2 opposed. Motion passed.	Y
4	900481420	Logan	Salgado	2026-06-10	10:00	Juliet Allen (JA), Beckett Barnett (BB), Amari	MM	Usually, these are longer....	JA motioned to readmit. MM seconded.	12 in favor, 0 opposed. Motion passed.	Y
5	900338956	Hayden	Mills	2026-06-10	10:20	Sawyer Ahmed (SA), Jaylen Blackwell (JB), V	SA	Minutes, minutes, minutes.	NZ motioned to readmit. GG seconded.	5 in favor, 4 opposed. Motion passed.	Y
6	900732688	Helen	Vance	2026-06-10	10:40	Kayson Chambers (KC), Dylan Freeman (DF),	DW	Here are some minutes.	AW motioned to deny the appeal. AS seconded.	10 in favor, 2 opposed. Motion passed.	N
7	900140433	Jesus	Stephenson	2026-06-10	11:00	Robert Atkinson (RA), Remington Hess (RH),	ER	These are the minutes.	RA motioned to readmit. ER seconded.	7 in favor, 2 opposed. Motion passed.	Y
8	900264085	Martina	Mcbride	2026-06-10	11:20	Ariella Ashley (AA), Paislee Bowers (PB), Aza	AB	More minutes.	FR motioned to readmit. AA seconded.	7 in favor, 2 opposed. Motion passed.	Y
9	900386328	Izabella	Dudley	2026-06-10	11:40	Jaden Bautista (JB), Eloise Guerra (EG), Geor	EG	Usually, these are longer....	JB motioned to readmit. RM seconded.	9 in favor, 1 opposed. Motion passed.	Y
10	900591515	Annalise	Hebert	2026-06-10	13:00	Scarlet Beltran (SB), Maisie Fields (MF), Octa	LH	Minutes, minutes, minutes.	ES motioned to readmit. QG seconded.	5 in favor, 4 opposed. Motion passed.	
11											
12											

Figure 7.3.2: Sample Spreadsheet with Errors

You may need to specify that all cells have text data in order for the `Date` and `Time` fields to show up appropriately. You can do this by selecting all of the cells you plan to use and then clicking on `Format` from the toolbar at the top of the spreadsheet window. The seventh option down will be `Cells` which will bring up a dialog window allowing you to specify the type of data in the cells.

Once you have all of the records entered, you need to save the spreadsheet in the correct format. From the toolbar, select `File` and then `Save As`... from the drop down menu. In the top toolbar, type in a name for the spreadsheet. You can also select what directory to save the spreadsheet in as well. Be sure to note where it is saved so that you can easily navigate back to it. Most importantly, in the bottom right–hand–side choose `Text CSV (.csv)` from the `All Formats` drop down menu.

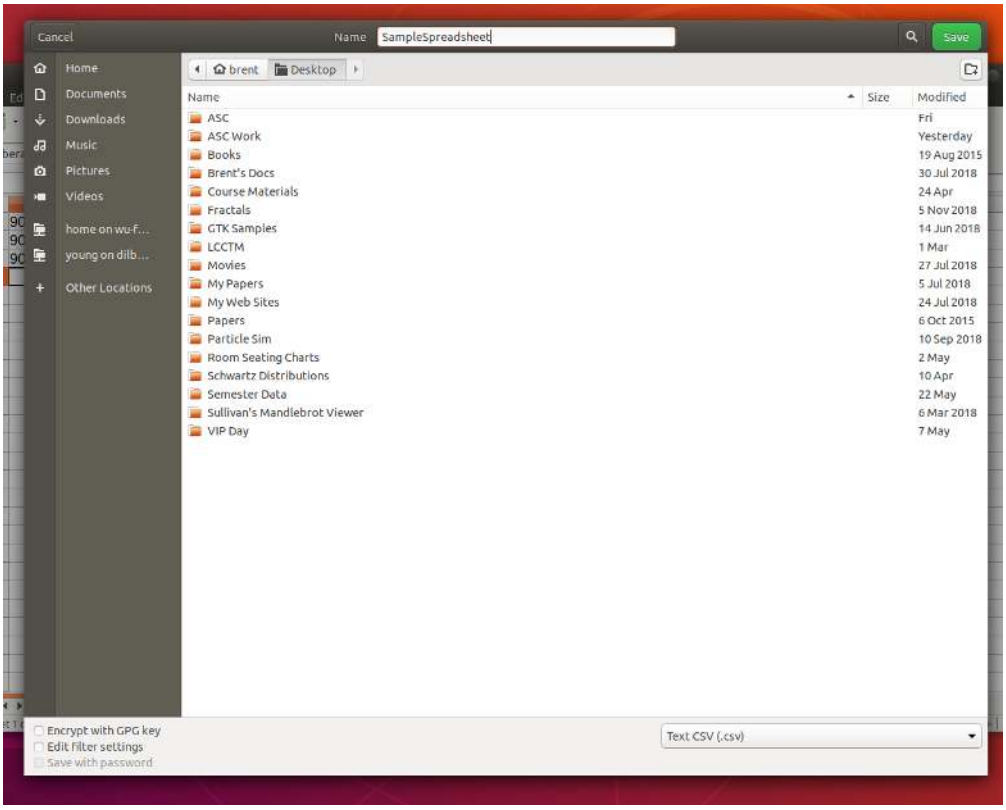


Figure 7.3.3: Spreadsheet `Save As` Settings

You should get a warning message informing you that certain formatting options may be lost if you use the CSV format. Choose `Use Text CSV Format`. This should bring up a dialog asking you how you would like the data to be exported. The options should be set as

- `Character set: Unicode (UTF-8)`
- `Field delimiter: {Tab}`
- `String delimiter: ''`

For the other options, you should only need to select `Save cell content as shown`.

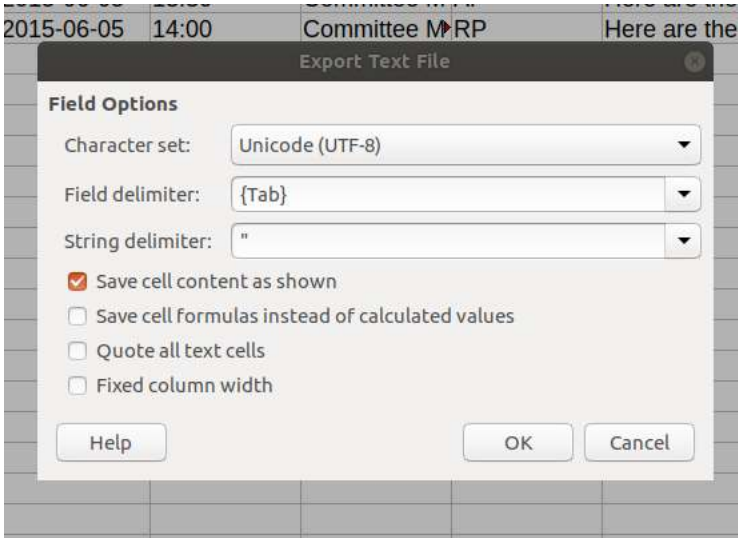


Figure 7.3.4: CSV Options

Selecting `Add Entries from File` from the main database control window will first bring up a dialogue reminding the user of the most important formatting details necessary to successfully import entries to the ASC Database. Clicking `Proceed` will bring up a dialogue box prompting the user to select the desired csv file.

Once `Open` is selected, the program will begin adding entries from the csv file to the database. Once this process is complete, a window will appear summarizing the results. Clicking `View Errors` will display a tabbed window

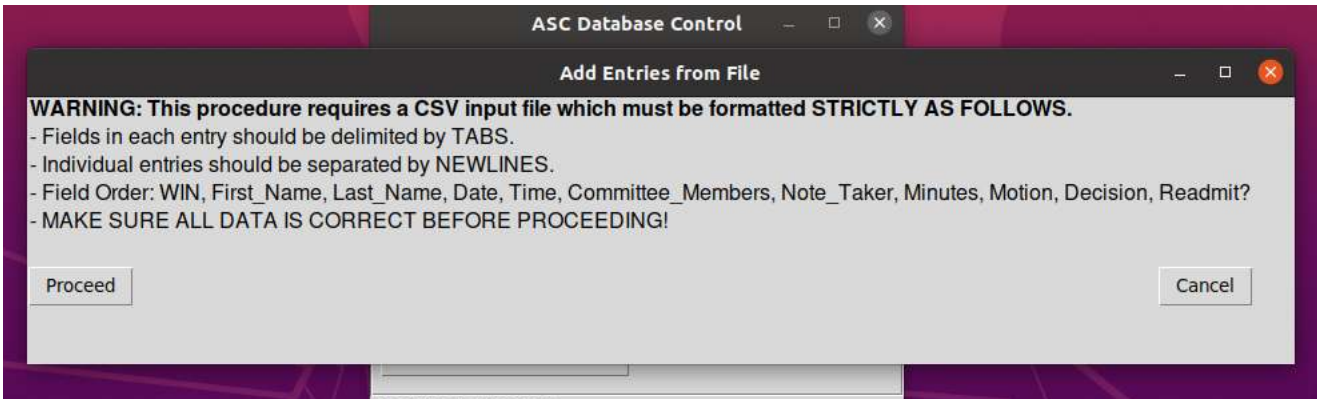


Figure 7.3.5: CSV File Formatting Reminder

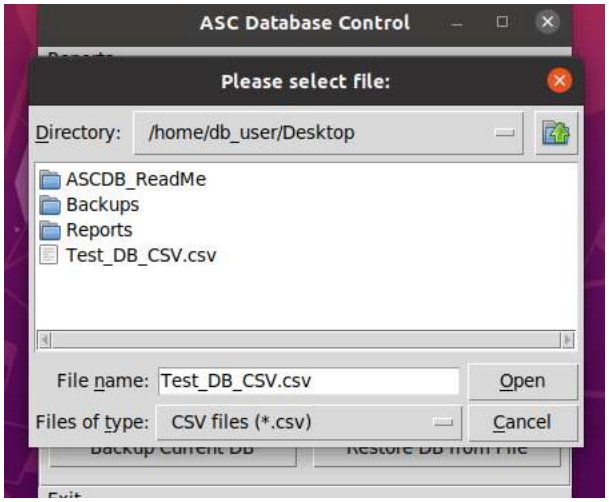


Figure 7.3.6: File Prompt

with each rejected entry getting its own tab detailing the reason for that decision. Formatting errors should be displayed fairly unambiguously. Errors reported by MySQL itself will also be displayed (with appropriate error codes which should allow the user to search the internet for advice).

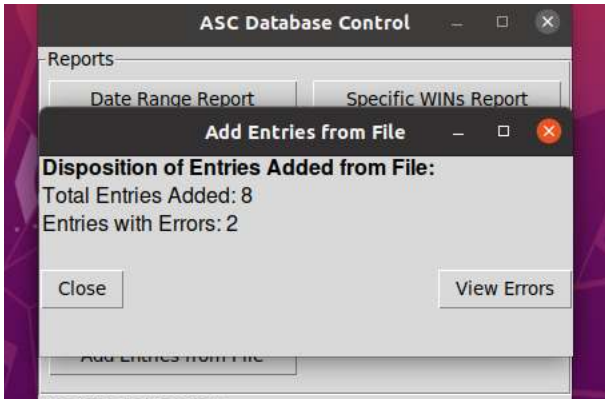


Figure 7.3.7: Add Entries from File Result Notification

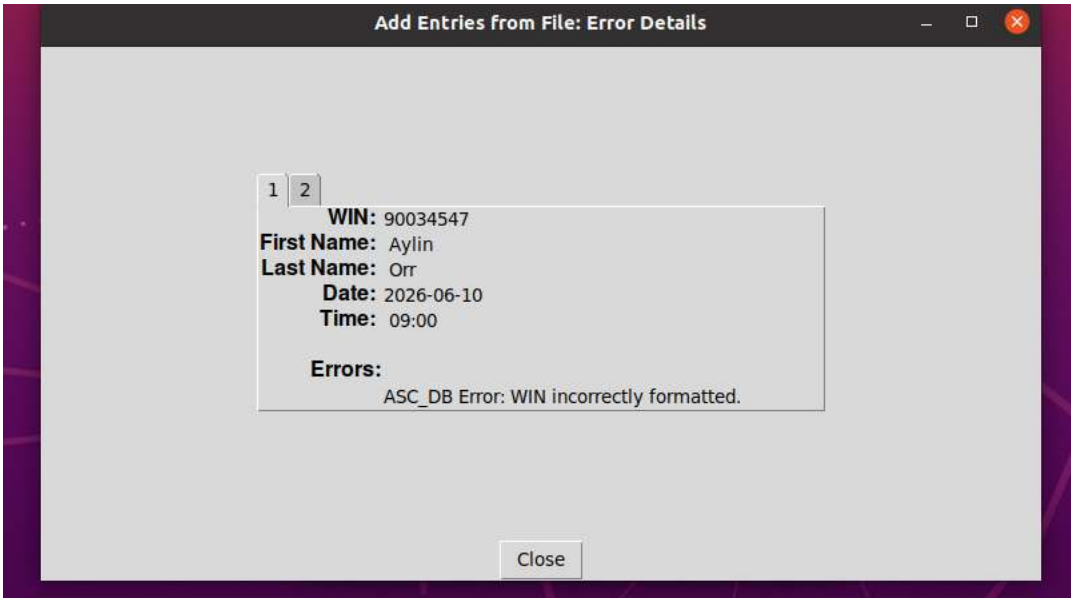


Figure 7.3.8: Add Entries from File Error Details

Hopefully, the user can make the appropriate corrections to the csv file with the given information and try the process again. Any duplicate entries will be ignored by the addition process (and an appropriate error message will be displayed to that effect). Again, verifying that the total number of

entries in the database has increased as expected is an easy way to double check that all new entries have been successfully added.

7.4 Delete Entries by WIN

Selecting `Delete Entries by WIN` from the main control window allows the user to remove all entries from the database associated to a given set of WINs. Be aware that this action is irreversible unless a backup of the current database has been created.

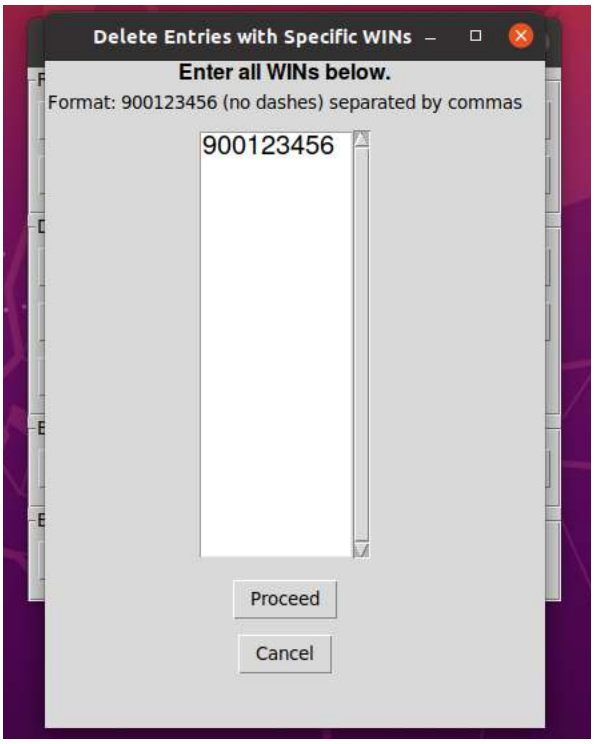


Figure 7.4.1: WIN Entry Dialogue

Once selected, the main control window will create a dialogue box prompting the user for a list of WINs. Each WIN should be entered *without dashes*, and a single comma should separate each WIN (no comma is necessary after the last WIN). *DO NOT PRESS ENTER AFTER ANY WIN.* The window has been sized to make each WIN appear on its own line for ease of reading.

Once all desired WINS have been entered, selecting `Proceed` will bring up a dialogue asking the user to confirm the deletion. Selecting `Proceed` a

Figure 7.4.2: Delete by WIN Warning

second time will execute the deletion process. A final dialogue will appear reporting the total number of entries deleted from the database.

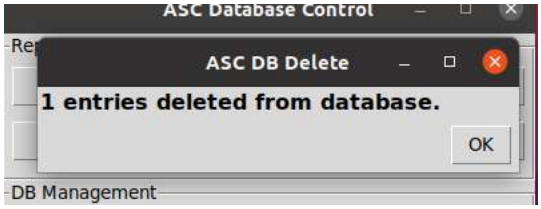


Figure 7.4.3: Delete by WIN Final Notification

7.5 Delete Entries by Date

As the name suggests, the menu option `Delete Entries by Date` allows the user to remove all entries between two given dates from the database. Be aware that this action is irreversible unless a backup of the current database has been created.

Once selected, the main control window will create a dialogue box prompting the user for the two dates. Notice that dates must be entered in the format `YYYY-MM-DD`. The program has been designed to catch most common errors (e.g. dates in the wrong order, day out of range of the given month, etc.) but care should be taken.

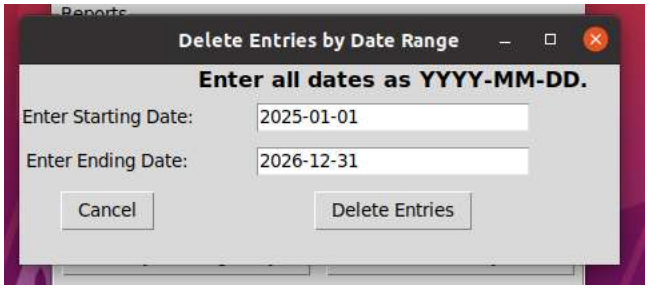


Figure 7.5.1: Delete by Date Prompt

Once the appropriate range of dates has been entered, a second dialogue will be created asking the user to verify that the desired dates are correct and warning the user that all entries in that date range will be permanently deleted from the database. Pressing `Proceed` will execute the deletion. A

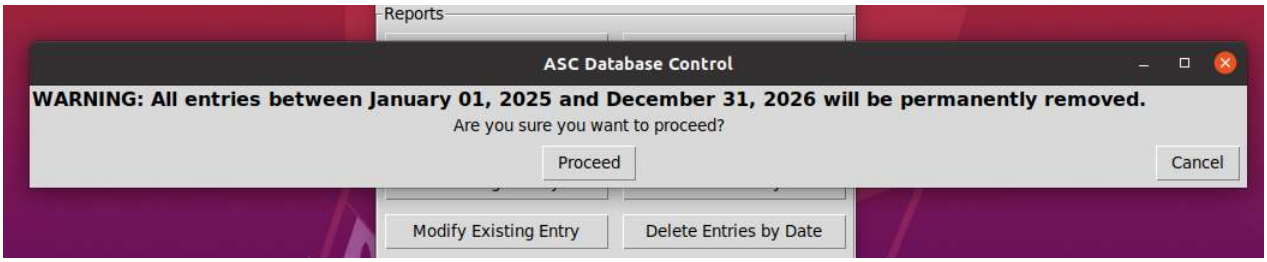


Figure 7.5.2: Delete by Date Warning

final dialogue will appear reporting the total number of entries deleted from the database.

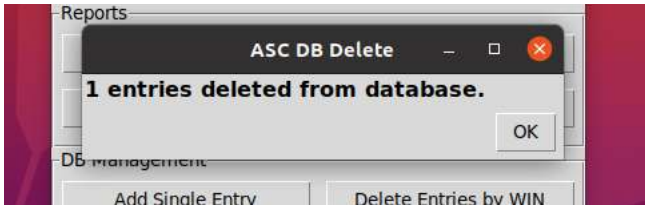


Figure 7.5.3: Delete by Date Final Notification

Chapter 8

Backup and Restore

8.1 Backup Current DB

The ASC Database should be backed up regularly. In particular, it is recommended that the database be backed up after any significant addition of entries.

As implemented, the backup protocol generates a MySQL dump file which essentially contains a series of procedures which is capable of fully recreating the database. A copy of the resulting backup file should be stored separately from the machine housing the database in case of some accident which compromises its functionality.

Clicking on `Backup Current DB` from the main control window will bring up a prompt asking the user to select a directory for the backup file to be stored. There is a folder on the desktop appropriately named `Backups` for storing these files.

Once the desired directory has been selected, a MySQL dump file will be created. The name of the file will be in the form

`AppealsDB_BackUp_date_time.sql`

where `date` will be replaced by a date stamp in the form YYYY–MM–DD and `time` will be a time stamp in 24–hour format. Once the backup has been created, a dialogue will appear confirming the name and location of the file.

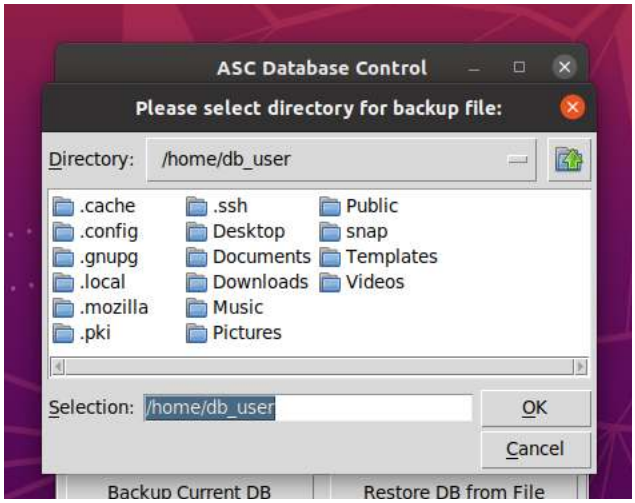


Figure 8.1.1: Back Up Directory Prompt

8.2 Restore DB from File

Should entries from the ASC Database be accidentally erased, it is possible to restore the database to an earlier state from a backup file. Be aware that

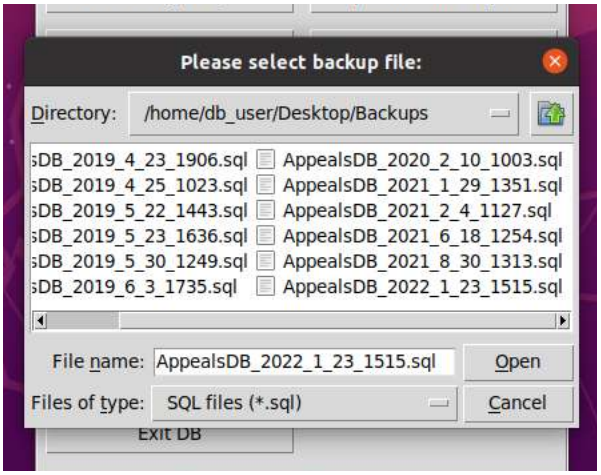


Figure 8.2.1: Restore DB Directory Prompt

the *entire database* will be replaced by this action. Once done, this action is

irreversible unless a backup file of the current database has been made. Make sure that restoring the database to an earlier state does not cause further entries to be deleted in the process!

To begin the restoration process, select `Restore DB from File` from the main control window. A dialogue box prompting the user to select the appropriate backup file will appear. The selected file must be an SQL dump file produced by the backup procedure in the preceding section. Once the desired backup file has been selected, a dialogue will appear warning the user that proceeding with the restore process will completely replace the existing database. Once the user selectes `Proceed`, the backup file will completely

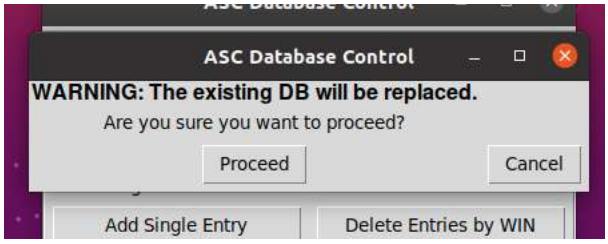


Figure 8.2.2: Restore DB Warning

erase the existing table of appeal records, create an empty table, and populate the table with the entries stored in the backup file.